

Madhulika Tirupatipanya

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Madhulika Tirupatipanya (<https://www.linkedin.com/in/madhulika-tirupatipanya-baa5a797/?skipRedirect=true>)

SUMMARY

Automation and Solution Design professional with 8+ years of experience designing and delivering technology solutions across banking, insurance and large-scale enterprise operations.

Specializes in converting complex, manually intensive business processes into structured, scalable and controlled digital workflows.

Experienced in owning the solution lifecycle from process discovery and stakeholder discussions through feasibility assessment, solution architecture, development, integration, testing, deployment, documentation and operational handover.

Combines strong technical depth across Microsoft Power Platform, VBA, SQL, Power BI, SharePoint, application development, APIs, Snowflake, dbt and Python with a strong understanding of business processes and operational constraints.

Known for identifying process loopholes, simplifying complex workflows, reducing manual intervention and designing solutions that work within real-world access, security and technology limitations.

EDUCATION

BITS Pilani

Post Graduate Programme in Artificial Intelligence & Machine Learning

2025 – 2026 | *In Progress*

JNTU

B.Tech in Aeronautical Engineering, Bachelor's Degree

2017

EXPERIENCE

Assistant Vice President

Wells Fargo

May 2024 – Present

Hyderabad, India

Automation & Solution Design

- Designed and delivered end-to-end automation solutions supporting commercial banking processes, working across business requirements, data extraction, business rules, reporting and workflow distribution.
- Led the solution design for a complex Loan IQ WIP process, involving identification of outstanding deals, deal ownership, group/manager mapping and distribution of actionable reports.
- Designed the technical flow across multiple components: Loan IQ → SQL → Web Portal → Raw Data → VBA Processing → Business Rules → WIP Report → Commercial Banking Distribution.
- Developed the SQL layer to identify relevant Loan IQ transactions and outstanding deals.
- Designed VBA processing architecture to reconcile SQL-derived deal information with raw portal data and apply complex business rules.
- Designed the ownership-mapping logic required to identify the appropriate deal administrator, group and management structure.
- Extended the solution into Xceptor, demonstrating the ability to adapt process architecture across different automation platforms.
- Designed a connected Power Apps + SharePoint + Power BI + Power Automate solution for operational tracking and reporting.
- Designed the process architecture where Power Apps acts as the user interface, SharePoint serves as the operational data layer, Power BI provides analytics and Power Automate orchestrates refresh and communication.
- Designed automated Power BI reporting workflows triggered by SharePoint events.

- Delivered 50+ VBA automations across operational processes, integrating report generation, SharePoint updates, shared-drive storage and automated email distribution.
- Worked directly with stakeholders and SMEs to identify process loopholes and translate operational requirements into feasible technical solutions.
- Designed solutions while minimizing dependency on privileged access and working within enterprise access restrictions.
- Served as SME across multiple processes and created simplified SOPs to standardize execution and reduce dependency on individual process knowledge.

Senior Software Engineer

February 2021 – May 2024

Accenture

India

- Designed, developed and enhanced enterprise automation applications across multiple client environments.
- Owned enhancements across 9 VBA applications, delivering more than 30 project enhancements.
- Supported complex insurance control processes involving extraction and consolidation of information from multiple databases.
- Worked directly with stakeholders to understand business requirements and identify technical solutions.
- Supported enterprise migration initiatives from on-premises infrastructure to GCP.
- Developed features for next-generation VB.NET applications.
- Designed integrations between C# applications and SharePoint using Microsoft Graph API and JavaScript.
- Supported a Siemens data platform migration involving dbt and Snowflake.
- Designed data-processing flows involving S3 ingestion, Snowflake staging, transformation views and final target tables.
- Worked across application, automation and data layers, providing a broad understanding of enterprise solution dependencies.

Quality Analyst

October 2017 – February 2021

Amazon

India

- Identified a repetitive catalog validation activity and designed a JavaScript-based solution to automatically highlight required validation fields.
- Improved the usability and efficiency of the quality validation process.
- Worked within high-volume operational processes with strict quality and accuracy requirements.

TECHNICAL SKILLS

End-to-End Process Solution Design

Process Discovery & Requirement Analysis

Automation Opportunity Identification

Technical Feasibility Assessment

Workflow Architecture

Business Rule Design

Data Flow & Integration Design

Low-Code Application Architecture

Reporting & Analytics Architecture

Process Controls

Stakeholder Management

SME Leadership

Process Standardization

SOP Development

Production Deployment & Support

Continuous Process Improvement

Power Apps

Power Automate

Power BI

Copilot
VBA
Excel Automation
Workflow Automation
SQL
Oracle SQL
Snowflake
dbt
ETL/ELT
Python
C#
VB.NET
.NET
JavaScript
HTML
REST APIs
Microsoft Graph API
SharePoint
Microsoft 365
S3
GCP
Machine Learning
Deep Learning
NLP
Generative AI

CERTIFICATIONS

Microsoft Azure DevOps Engineer Expert — AZ-400
Microsoft Azure Administrator — AZ-104
Microsoft Azure Fundamentals — AZ-900
Oracle Database SQL Certified Associate

LEADERSHIP & BUSINESS CONTRIBUTION

Stakeholder & Requirement Management: Engage with business users and SMEs to understand the actual process rather than simply automating individual manual steps.

Process Optimization: Identify unnecessary manual intervention, duplicated effort, process gaps and control weaknesses before selecting the technical solution.

Solution Design: Evaluate available technologies and design practical solutions based on data availability, access restrictions, existing infrastructure, user requirements and maintainability.

SME & Documentation: Create standardized SOPs, process documentation and user guidance to make automated processes sustainable beyond the original developer.